

# LaborCoin Whitepaper

## Abstract

LaborCoin is a decentralized governance system designed to enable collective decision-making and resource allocation for the working class. It combines a non-transferable voting token, DAO-based governance, and controlled execution through dedicated smart contracts to create a transparent, accountable, and scalable coordination mechanism.

The system enforces a one-person, one-vote model and restricts execution to predefined actions, minimizing attack surface while preserving flexibility through governance.

In addition, LaborCoin integrates a bonding curve exchange mechanism that enables continuous liquidity, deterministic pricing, and treasury accumulation to support striking workers and collective action.

## 1. Introduction

Modern economic systems concentrate decision-making power in centralized institutions, limiting the ability of workers to coordinate and act collectively. While blockchain technology enables decentralized coordination, many implementations prioritize speculation over governance.

LaborCoin addresses this gap by providing:

- A governance-first architecture
- Equalized voting power
- Transparent, on-chain decision execution
- Controlled and auditable action pathways
- A treasury-building mechanism tied to real economic participation

The goal is not to create a speculative asset, but a functional coordination system capable of sustaining worker-led movements.

## 2. System Overview

- LABR (primary token)
- LABRV (voting token)
- DAO governance (Aragon)
- Executor contracts
- Bonding curve exchange

Participants → LABR → LABRV → Proposals → DAO → Executors → Treasury → Worker Support

This architecture ensures decisions are enforced through constrained execution while maintaining economic backing through the bonding curve.

### **3. Governance Model**

Voting Mechanism

1 LABRV per participant  
Non-transferable (soulbound)  
Equal voting weight

Prevents capital-based governance capture and ensures democratic participation.

Thresholds

25% participation  
67% approval

High thresholds ensure legitimacy and prevent minority control.

### **4. Token Design**

Contract Address:  
0x460DD873A1D2a41e77410B125cD3027C5FE2f78

Total Supply: 1,000,000,000  
Initial Supply: 100,000,000

Emission Model

50,000,000 LABR released per tranche  
1:1 release ratio up to 50% of total supply  
1:0.8 release ratio beyond 50%

This structure encourages early participation while gradually introducing scarcity.

Constraints

Transaction Limit: 5,000 LABR per transaction  
Wallet Limit: 10,000 LABR per wallet  
Cooldown: 12 hours between transactions

These constraints are enforced at the contract level to prevent accumulation of disproportionate influence and reduce volatility.

## 5. Bonding Curve Exchange

Exchange Contract:

0xD0692ec758bb852421B702B187b6439f74f8Bf3b

Pricing Model

Quadratic bonding curve with smooth price progression

Price increases as total tokens sold increases

Approximate range: \$1 → ~\$15

Economic Behavior

Early participants enter at lower prices, incentivizing initial adoption.

As participation grows, price increases smoothly rather than abruptly.

This discourages speculation-driven spikes and instead rewards sustained participation.

Treasury Model

All purchases deposit POL into the contract treasury

Sales withdraw from treasury liquidity

10% of each purchase is routed to the DAO treasury

This creates a direct link between participation and available support funds.

## 6. Execution Architecture

DAO → Executor → Contract

Executors

PauseExecutor → controls trading state

TreasuryExecutor → distributes funds

MintExecutor → mints voting tokens

Execution is constrained to prevent arbitrary control and reduce attack surface.

## 7. Deployment

Network: Polygon

LaborCoin (LABR):

0x460DD873A1D2a41e77410B125cD3027C5FE2f78

LaborCoin Exchange:

0xD0692ec758bb852421B702B187b6439f74f8Bf3b

LaborCoin DAO:

0x0C2e5679153593b82a84eAB5CA90895BB291Cec4

DAO Registration (v2)

0xFFc3499A71b806C3919f4B54D236b151cFdCB453

LaborVote (LABRV v6):

0x113579220515cd59b884Ea2379b4C369025246e2

DAO Governance (v4):

0x69b3317D3D742AFa6cd770F7cA29Eeeacdda073D

Pause Executor (v2):

0x94c83d82Bd04947C3eC426b99C316d49dA977af2

## **8. Security Model**

Restricted execution pathways

Immutable economic parameters

Non-transferable governance

High consensus thresholds

Additional Protections

Oracle-based pricing validation

Slippage protection

Circuit breaker (pause mechanism)

Reentrancy protection

## **9. System Invariants**

The following conditions are designed to always hold:

No participant can obtain more than one voting token

Governance decisions cannot execute arbitrary contract logic

Treasury withdrawals cannot exceed available balance

Token supply cannot exceed maximum defined limits

Pricing remains deterministic and independent of external markets

These invariants define the core guarantees of the system.

## **10. Threat Model**

LaborCoin assumes adversarial behavior as a baseline condition.

### **Adversarial Actors**

Wealthy participants attempting to accumulate influence  
Coordinated voting blocs attempting governance capture  
Smart contract exploiters targeting liquidity  
External actors manipulating price inputs

### **Attack Surfaces**

Bonding curve pricing  
Treasury liquidity  
DAO participation thresholds  
Executor permissions  
Oracle feeds

## **11. Economic Attack Vectors**

### **Whale Accumulation**

Mitigation: Wallet caps, transaction limits, and non-transferable voting tokens.

### **Liquidity Drain Attacks**

Mitigation: Increasing price curve, cooldown enforcement, treasury constraints.

### **Speculative Volatility**

Mitigation: Deterministic pricing independent of external markets.

## **12. Governance Risks**

### **Low Participation**

Mitigation: High quorum requirements.

### **Coordination Attacks**

Mitigation: 67% approval threshold.

Voter Apathy

Acknowledged as a social limitation inherent to democratic systems.

### **13. Failure Scenarios**

Oracle Failure

If price oracle data becomes invalid, transactions relying on price inputs may fail or pause.

Treasury Depletion

If treasury liquidity is insufficient, sell transactions may fail until replenished.

Low Participation

Governance may stall if participation thresholds are not met.

Contract Pause

The system may enter a paused state under emergency conditions to prevent further damage.

### **14. System Assumptions**

Participants engage in governance

Worker solidarity can be coordinated

Blockchain infrastructure remains secure

Oracle pricing remains reliable

LaborCoin minimizes trust in individuals but depends on collective participation.

### **15. Use Cases**

Strike funding

Mutual aid

Collective bargaining

Resource allocation

### **16. Conclusion**

LaborCoin shifts blockchain systems from speculation to coordination infrastructure. It combines governance, economics, and execution into a unified system capable of supporting real-world collective action.